1 Introduction to cloud native

This chapter covers

- What the cloud and cloud computing model are
- The definition of cloud native
- Characteristics of cloud native applications
- Culture and practices supporting cloud native
- When and why you might consider the cloud native approach
- Topologies and architectures for cloud native applications

Cloud native applications are highly distributed systems that live in the cloud and are resilient to change. Systems are made up of several services that communicate through a network and are deployed in a dynamic environment where everything keeps changing.

Before diving into the technologies, it's fundamental to define what cloud native is. Like other buzzwords in our field (such as *agile*, *DevOps*, or *microservices*), *cloud native* is sometimes misunderstood and can be a source of confusion because it means different things to different people.

In this chapter, I'll provide you with the conceptual tools you'll need for the rest of the book. I'll start by defining what cloud native means and what it takes for an application to be identified as such. I'll explain the properties of cloud native applications, examine the characteristics of the cloud computing model, and discuss when and why you might want to move to the cloud. I'll also present some fundamental concepts of cloud native topologies and architectures. Figure 1.1 shows an overview of the different elements I'll